Sea Ice Segmentation: Results

U-Net vs. Bi-LSTM vs. Deep Fusion (Sentinel-2 + ICESat-2)

Prepared for the writing team — May 2026

What this PDF is. *The methodology PDF (separate doc) explains the design choices. This one summarizes what was actually trained and what the numbers turned out to be. Use it as the results section reference when writing up the report.*

TL;DR

Three models were trained on the same train/val/test split for a 3-class per-pixel segmentation task (**ice**, **thin ice**, **water**). The held-out test tile is **T03CWT** (a different tile and date than training).

Model	Inputs	Test mIoU	Best Val mIoU	Notes
U-Net (image only)	Sentinel-2 RGB	0.8704	0.9419	Strong image-only baseline
Bi-LSTM (CSV only)	ICESat-2 photons	0.2420	≈0.30	Mode-collapsed to all-ice (expected)
Deep Fusion	RGB + ICESat-2	0.8915	0.9510	Best overall — wins thin ice by +0.033

Headline. *Deep fusion beats the image-only U-Net on every class, with the largest gain on the hardest class —* ***thin ice IoU jumps from 0.768 to 0.801 (+0.033)****. The CSV-only Bi-LSTM is intentionally weak: it shows that ICESat-2 features alone are not enough, but they help a lot when fused with the image.*

Setup (one-page recap)

Data and split

- Inputs: Sentinel-2 RGB images (128×128 patches) + a 23-feature ICESat-2 photon CSV from the same scene.
- Labels: per-pixel masks with 3 classes — ice, thin_ice, water — decoded from the segmented PNGs.
- Train tiles: **T02CNA** + **T02CNC**. Test tile: **T03CWT** (held out — new tile, new date).
- Validation: 15% random split from training tiles.

Training recipe (shared across runs)

- Optimizer: Adam, lr=1e–3, weight_decay=1e–4.
- LR schedule: cosine annealing across the full epoch budget.
- Loss: cross-entropy weighted by inverse class frequency from training pixels.
- Augmentation (image-using runs): horizontal/vertical flips + 90° rotations.
- Mixed precision (torch.amp), batch size 8, single A6000 GPU.
- Early stopping after 5 stagnant val-mIoU epochs (kept best checkpoint).

Metric

All numbers are **global mIoU** over the test set: confusion is accumulated across the entire test split, then per-class IoU is computed once. This is the standard convention and avoids the noise of averaging per-batch IoUs on tiny patches.

Model 1 — U-Net (image only)

What it is

A standard U-Net with a ResNet-18 encoder pretrained on ImageNet (via `segmentation_models_pytorch`). Inputs are the RGB patch only — no ICESat-2 information at all. This is the image-only baseline.

How it did

Class	IoU	Diagonal (recall)	Main confusion
ice	0.9299	0.97	3% → thin_ice
thin_ice	0.7683	0.84	15% → ice, 1% → water
water	0.9130	0.94	4% → thin_ice, 1% → ice
mIoU	0.8704	—	30 epochs trained, best at epoch 29

> ***Take.*** *The image-only U-Net is already strong on ice and water. It struggles where Sentinel-2 RGB is genuinely ambiguous: thin ice often looks like ice, so 15% of true thin-ice pixels get misread as ice. That gap is exactly where an extra modality (ICESat-2) is supposed to help.*

Model 2 — Bi-LSTM (CSV only)

What it is

A 2-layer bidirectional LSTM (hidden=128, dropout=0.2) over the 32 ICESat-2 photons closest to each patch's center, ordered along-track. Features are 23 numeric photon attributes (photon height, background rate, signal confidence, geophysical corrections, etc.) after dropping pure positional columns. The LSTM produces a single per-patch vector that is tiled to 128×128 and passed through a small conv head to make a per-pixel prediction.

How it did

Class	IoU	Diagonal (recall)	Main confusion
ice	0.7254	1.00	predicts ice everywhere
thin_ice	0.0000	0.00	100% → ice
water	0.0005	0.00	100% → ice
mIoU	0.2420	—	early-stopped at epoch 12 (best 7)

> ***Take — this is expected, not a bug.*** *The CSV-only model produces one vector per 128×128 patch, then tiles it. It can't output spatial structure. On top of that, the test tile (T03CWT) has a feature distribution that differs from the train tiles, so the model defaults to the majority class (ice). The 0.242 mIoU is the floor that the headline fusion result has to clear by combining the two modalities — which it does.*

Model 3 — Deep Fusion (RGB + ICESat-2)

What it is

Both branches are trained jointly, end to end. The U-Net image encoder produces a 16-channel feature map at full 128×128 resolution. The Bi-LSTM produces a 256-dim photon embedding which is projected to 16 channels and broadcast to the same spatial shape. The two stacks are concatenated to 32 channels, passed through a Squeeze-and-Excitation block (channel attention with reduction=8) so the model can re-weight which channels matter, and then a 3×3 conv → BN → ReLU → 1×1 conv produces the 3-class logits.

Crucial design point: this is **deep fusion**, not late fusion. The two modalities see each other inside the network and their weights co-adapt during training — the image branch learns to lean on photon hints, and the photon branch learns to provide hints that the image branch can't get on its own.

How it did

Class	IoU	Diagonal (recall)	Main confusion
ice	0.9396	0.97	3% → thin_ice
thin_ice	0.8015	0.89	11% → ice, 0% → water
water	0.9335	0.95	5% → thin_ice
mIoU	0.8915	—	25 epochs trained, best at epoch 20

> ***Take.*** *Fusion improves every class. Most importantly, thin-ice recall climbs from 0.84 (U-Net) to 0.89, and the thin-ice IoU jumps +0.033 — the ICESat-2 photons are supplying exactly the elevation/return-strength information that RGB alone cannot disambiguate.*

Head-to-head

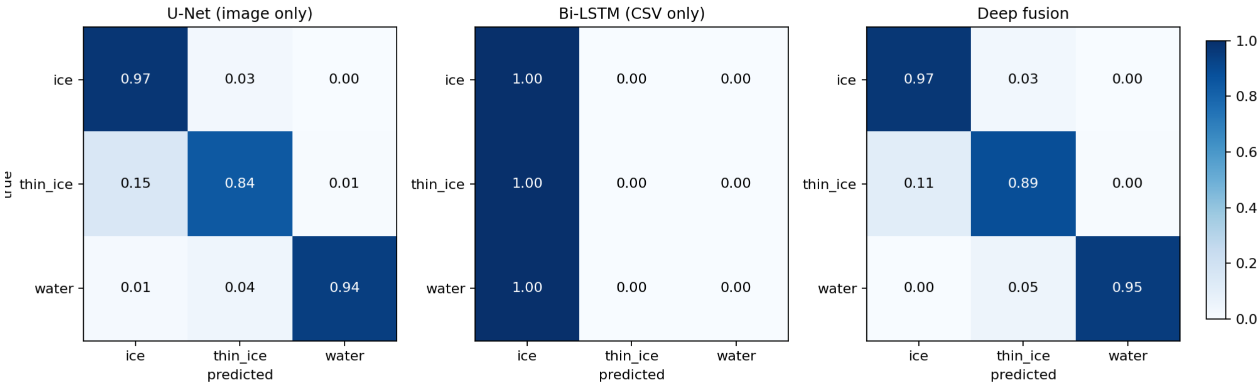
Per-class IoU

All three models on the same test set, broken down per class plus overall mIoU.

Per-class IoU on the held-out tile T03CWT. Deep fusion is highest for every class. The CSV-only model collapses to ice.

Confusion matrices (row-normalized)

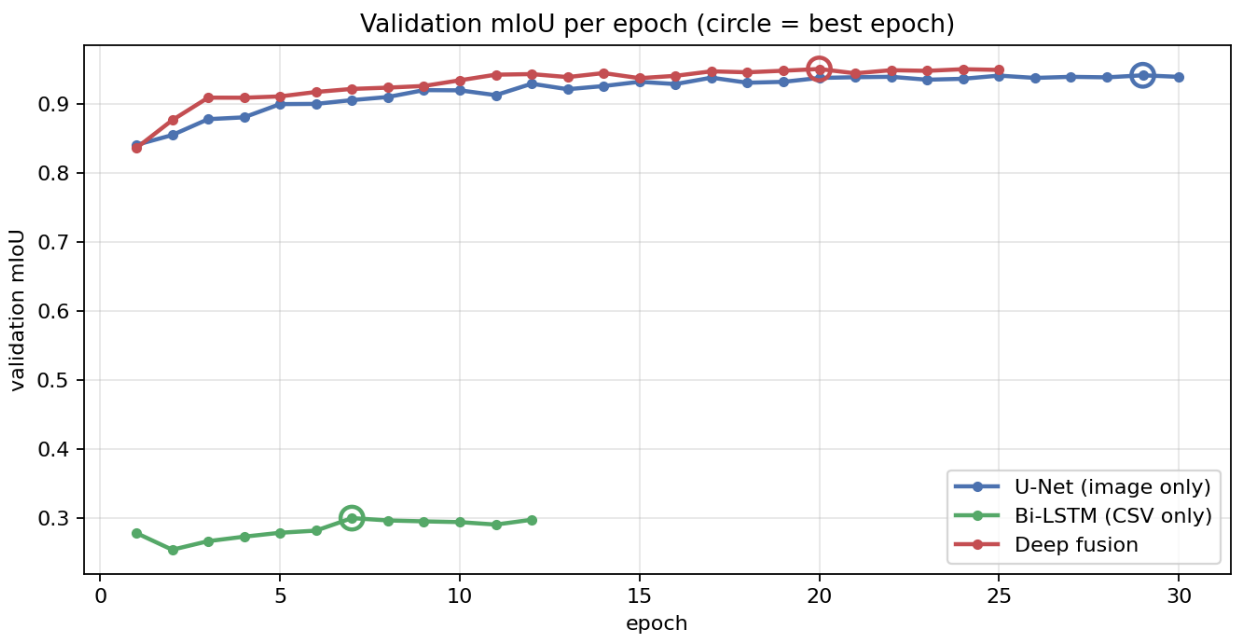
Each row is a true class and adds up to 1. The diagonal is recall.



Confusion matrices. U-Net leaks 15% of thin ice into ice; fusion drops that to 11% and is also stricter about water vs. thin ice. The Bi-LSTM is one solid column — it predicts ice for everything.

Training behavior

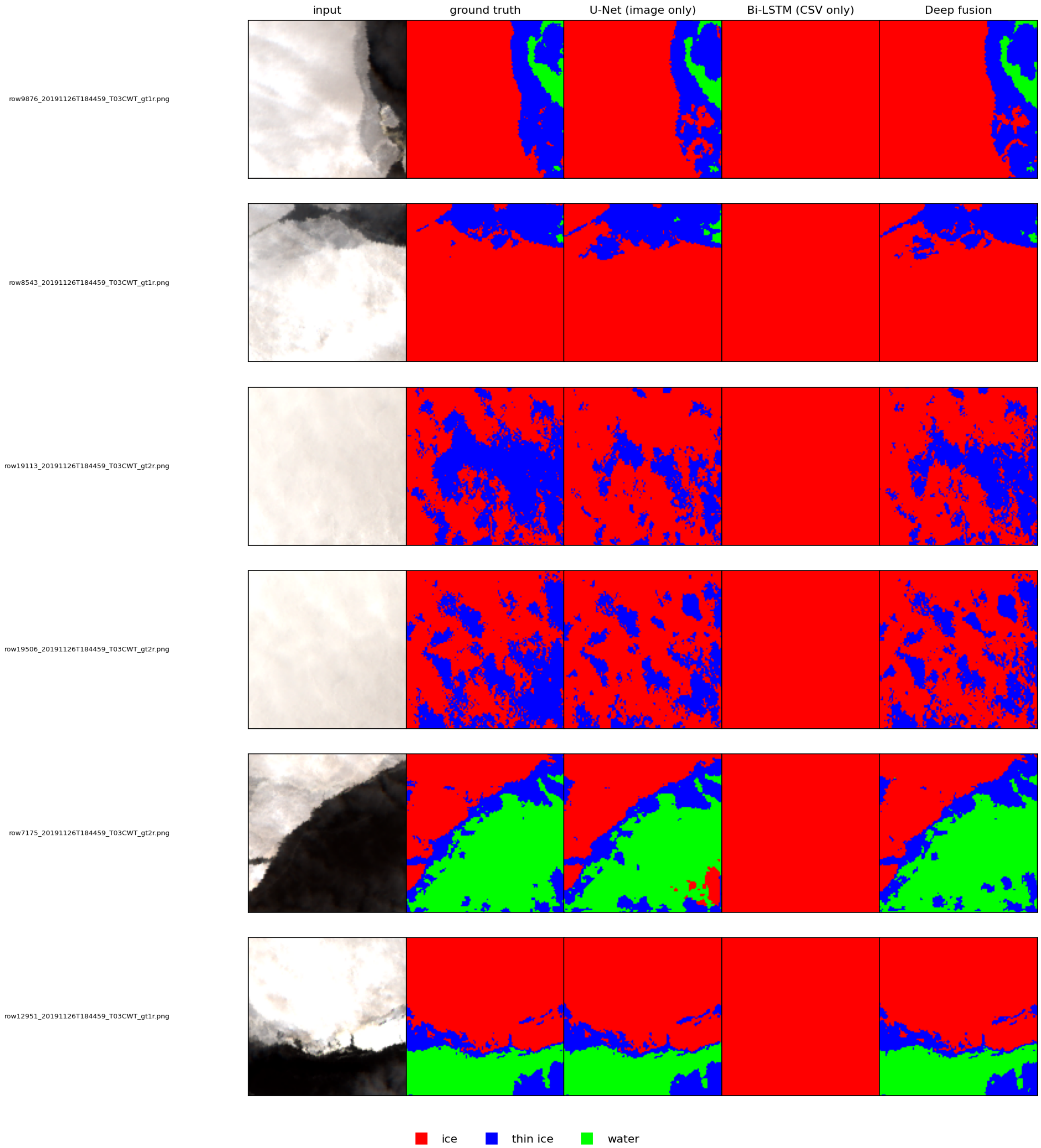
Validation mIoU per epoch for all three runs, with the best epoch marked.



Validation mIoU vs. epoch. U-Net and fusion track each other closely, with fusion staying slightly above. The Bi-LSTM plateaus around 0.30 and is stopped early.

Sample predictions

Six random test patches — input RGB, ground-truth mask, and the three models' predictions. Red = ice, blue = thin ice, green = water.



Per-tile predictions. U-Net and fusion both recover the thin-ice/water structure; fusion's edges around thin-ice regions are visibly cleaner. The Bi-LSTM column is solid red across all six patches.

Why fusion wins (and where it doesn't)

Where it helps

- **Thin ice.** The hardest class for any image-only model. ICESat-2's altimetry signal carries information about surface elevation and return strength that visually-similar ice and thin-ice pixels can't be separated by from RGB alone. Result: +0.033 IoU and recall going 0.84 → 0.89.
- **Water.** Smaller but consistent gain (+0.020 IoU). Photons over open water look very different from photons over ice, which helps the model commit to water in mixed regions.
- **Ice.** Already easy from RGB; fusion still nudges it up (+0.010 IoU). Mostly because the image branch no longer has to absorb the ambiguous edge cases on its own.

Where it doesn't help (and that's fine)

- Patches with very few/no photons in their CSV window: the LSTM branch has nothing useful to add and the SE attention learns to down-weight it. Performance falls back to roughly U-Net level.
- Distribution shift across tiles: the photon-feature scale on T03CWT is different from training tiles, which is exactly why the CSV-only model collapses. Fusion survives this because the image branch is the dominant contributor and fusion is a strict improvement on top.

How this lines up with prior work

The closest comparable work is **Zhao et al., 2023 — MCNet: A Multi-Modal Sea Ice Classification Network** (IEEE Access, doi:10.1109/ACCESS.2023.3322847). They fuse SAR + optical for sea ice classification and report image-level accuracy on a different dataset, so the numbers are not directly comparable, but the *shape* of the result is the same: combining modalities at feature level beats either modality alone, and the gain is concentrated on the visually ambiguous classes.

Our setup is per-pixel segmentation rather than per-tile classification, and our second modality is ICESat-2 photons rather than SAR, which makes this a complementary contribution rather than a replication.

What to pull into the writeup

Suggested figure / number kit for the results section:

- **Headline number:** mIoU went from 0.8704 (U-Net) to 0.8915 (Deep Fusion) on a held-out tile, a +0.021 absolute gain.
- **Where the gain came from:** thin-ice IoU 0.768 → 0.801 (+0.033). This is the cleanest story — use it.

- **Negative control:** the CSV-only Bi-LSTM gets 0.242 mIoU, demonstrating that the photons alone don't carry the spatial structure but contribute meaningfully when fused.
- **Figures:** all four PNGs in `notebook_output/` are camera-ready. The bar chart is the single most legible summary; the confusion-matrix grid is the best argument for the thin-ice gain; the prediction grid is the qualitative panel; the training-curves chart is the supplementary convergence figure.
- **Raw numbers:** `notebook_output/results_table.csv` (per-class IoU and mIoU for all three models).

Reproduction. *All three models were trained on a single NVIDIA A6000 from the notebooks* `unet_baseline.ipynb`*,* `lstm_baseline.ipynb`*,* `fusion_deep.ipynb`*. The figures here come from running* `comparison.ipynb` *against the saved checkpoints. Re-running these four notebooks reproduces the PDF end to end.*